Branching, Inversion, and Meta-Operators

$$\vec{a}_k G_k \vec{a}_k \rightarrow \vec{a}_{(g)|k}$$

- Modal vector $\vec{a}_k$ is transformed by operator G_k, producing a new modal state $\vec{a}_{(g)|k}$.
- Inverses G_k^{-1} allow backtracking or resolving ambiguities—analogous to time reversal or undoing a transformation.

$$[\cdots \mathfrak{F}' \varepsilon \mathfrak{F}'] \rightarrow \mathfrak{F}'$$

- Meta-operator $\mathfrak{F}'$ closes the construction: if a chain "loops back" via identity (ε), it projects to a higher-level order.

Like a Feynman sum, but with branching, recursion, and closure at every level.